

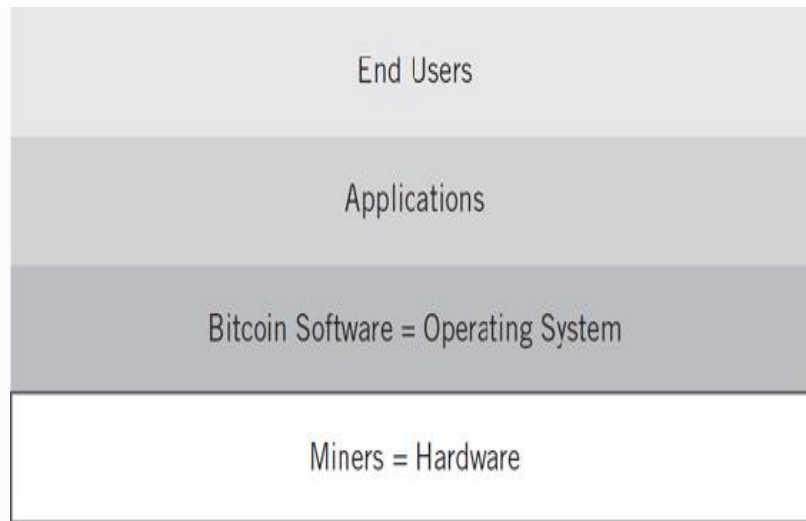


Figure 2.1 ■ Bitcoin as a stack of hardware, OS, applications, and end users

The miners that build Bitcoin's blockchain with the PoW process are the hardware, just as a MacBook Pro provides the hardware for a personal computer. That hardware runs an operating system (OS); in the case of Bitcoin, the operating system is the open-source software that facilitates everything described earlier. This software is developed by a volunteer group of developers, just as Linux, the operating system that underlies much of the cloud, is maintained by a volunteer group of developers. On top of this hardware and operating system combination are applications, just as Safari is an application that runs on an Apple operating system. The applications interface with the Bitcoin operating system, which pushes and pulls information to and from Bitcoin's blockchain as needed. Lastly, there are the end users that interface with the applications, and someday may have no concept of the hardware or software underneath because all they need to know is how to navigate the applications.